From Reactive to Systemized

An architectural precast turnaround — and what integrated production management is worth to a plant like yours

How a ~100-person architectural precast plant in Northern California — running the earliest generation of the IntraSync platform — turned integrated production management into $3.28M of annual value. Every figure is built from the plant's real revenue and downtime logs, with each modeling assumption stated in the methodology.

$3.28M	$2.42M	$858K
Estimated total annual value	Revenue enabled by recovered capacity	Hard cost savings per year

The operation

The plant is a ~100-person architectural precast producer in Northern California (~$30M annual revenue) — custom cladding, GFRC, and structural-architectural panels where finish quality and on-time delivery decide whether a job is profitable. It was here that IntraSync's founder built and deployed the first generation of the **CastLogic** platform directly on the plant floor at **Walters & Wolf Precast** — a PCI-Certified architectural precast and GFRC plant (PCI categories AA and G) in Fremont, CA. The figures below combine the plant's **real revenue and downtime logs** with conservative, industry-cited improvement rates — each one stated so you can substitute your own.

The challenge: the logged cost of running disconnected

Before an integrated system, the plant ran the way most precasters still do: scheduling in spreadsheets, estimating by hand, quality caught late, and maintenance handled only after something broke. The plant's own logs put a number on that fragmentation.

921 hrs*	$2,675	$15K	$341K
Unplanned downtime logged annually	Direct cost per downtime hour (idle labor + burden)	Production value at risk per hour	Annual clerical / admin salary load

**Plant-measured from real downtime logs. This is roughly 2.8× the large-plant industry average (~326 hrs/yr, Siemens 2024) — high, and defensible only because it is the plant's own recorded figure, not a benchmark.*

Against those logs, unplanned downtime alone cost **$2.46M a year** in direct idle labor and burden (921 hrs × $2,675) — before counting the $15K/hr of production those stoppages put at risk.

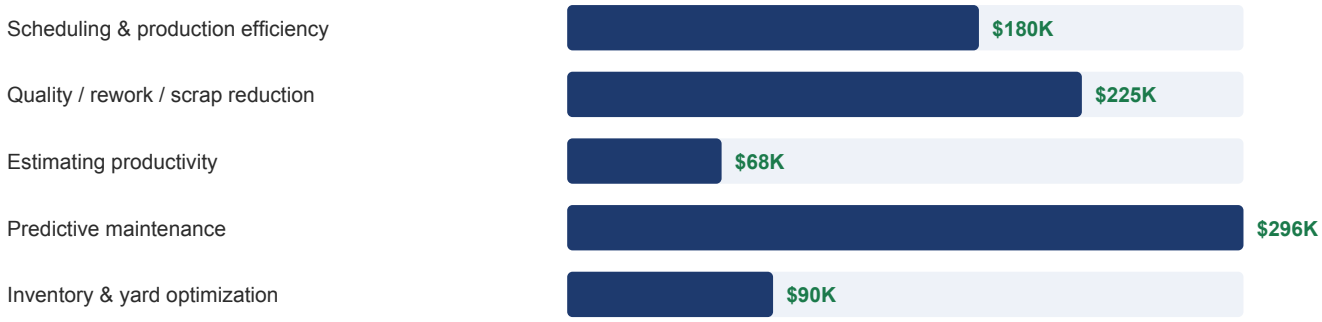
The solution: one connected system

Connecting the plant's core functions on one platform attacked the waste at its source. Five levers drove the hard savings; recovered uptime drove the revenue.

Scheduling & production	AI-assisted mold and labor sequencing replaced spreadsheet planning — fewer idle beds, fewer collisions, less downtime.
Quality & rework	Defects and finish rejections caught in-process instead of at ship — less scrapped product, fewer remakes.
Estimating	Structured, reusable estimates cut quote turnaround and freed senior estimators from re-keying.
Predictive maintenance	Condition-based servicing replaced run-to-failure, converting emergency stoppages into planned ones.
Inventory & yard	Real-time material and yard visibility trimmed carrying cost and lost-part hunts.

The return, part 1: hard cost savings you can bank

Lever	Annual savings
Scheduling & production efficiency	**$180K**
Quality / rework / scrap reduction	**$225K**
Estimating productivity	**$68K**
Predictive maintenance	**$296K**
Inventory & yard optimization	**$90K**
Total hard cost savings	**$858K**



The return, part 2: revenue the recovered uptime lets you earn

Removing 35% of logged downtime returns **322 production hours** a year. Valued at $15K/hr and discounted to a conservative **50% sold-through** (demand-limited, not automatic), that is **$2.42M in additional revenue** — an 8.1% uplift. At a 30% contribution margin, its profit contribution is ~$725K.

$3.28M total annual value

Hard cost savings ($858K) plus revenue enabled ($2.42M) **= $3.28M in total annual value.** Hard savings alone ($858K/yr ≈ $72K/mo) recover a typical ~$350K implementation in **4.9 months** — before the revenue uplift. (Industry-median ERP payback is ~16 months; this is faster because the savings are front-loaded operational, not deferred.)

How every number is derived

Figure	Derivation
Production value / hr	$30M revenue ÷ 2,000 production hrs (1 shift) = $15K
Downtime exposure	921 hrs × $2,675/hr = $2.46M [plant logs]
Scheduling & production	$30M × 20% prod-labor × 3% efficiency = $180K
Quality / rework	$30M × 3% scrap × 25% cut = $225K
Estimating	3 estimators × $90K × 25% time saved = $68K
Predictive maintenance	$2.46M × 40% failure-related × 30% cut = $296K
Inventory & yard	$30M × 10% inv × 20% carry × 15% cut = $90K
Revenue enabled	322 hrs recovered × $15K/hr × 50% sold = $2.42M
Total annual value	$858K savings + $2.42M revenue = $3.28M
Payback	$350K ÷ ($858K/12) = 4.9 months

Real (plant-measured): $30M revenue, 921 downtime hrs, $2,675/hr downtime cost, $341K clerical load. **Modeled (stated, adjustable):** 2,000 production hrs, 35% downtime reduction, 50% capacity sold-through, 30% margin, and the scrap/labor/inventory rates above.

Benchmarks: Siemens True Cost of Downtime 2024 (downtime reduction 30–50%); NPCA Precast Market Report 2022 ($253K revenue/employee); U.S. BLS OEWS NAICS 327000 (admin wages, corroborates $341K as conservative); Nucleus Research & Panorama Consulting (ERP payback context).

> *"I didn't design this from a whiteboard. I built the first version standing on the plant floor, watching where the hours and the panels actually disappeared. CastLogic is that same operational logic, rebuilt on modern tooling with AI in every module."*
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> **— Zachary Frye, Founder, IntraSync Industrial**

This is what we rebuilt CastLogic to do

CastLogic is the only ERP platform designed specifically for precast — 13 integrated modules, AI embedded throughout, from quote to cash. Give us your plant's real revenue and downtime numbers and we'll rebuild this model against **your** floor.

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Modeled from the operational experience of running the earlier-generation platform at a ~100-person architectural precast plant; not an audited CastLogic deployment. Walters & Wolf is referenced solely as the site of that earlier system. Revenue ($30M), downtime hours (921), downtime cost ($2,675/hr) and clerical load ($341K) are plant-measured; improvement rates are stated, conservative, and industry-cited. Revenue-enabled figures are top-line and demand-dependent. Actual results vary with current operations, scope, and implementation.